$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \qquad tr(A) = 0, \quad Det(A) = -2$$

$\lambda_1 = \sqrt{2}$ ; $\lambda_2 = -\sqrt{2}$

Now, $A^2 = 2I \Rightarrow A^3 = 2A \Rightarrow A^4 = 4I$

$\Rightarrow A^{13} = 64A$

$\therefore$ Eigenvalues of $A^{13} = 64$ (eigenvalues of $A$)